

PREDICTION TASK  <u>Type of task:</u> Regression model <u>Entity:</u> a used vehicle's specifications including model, year, condition, mileage <u>Outcomes:</u> Decision support of buy/pass, and outputs the expected profit, max purchase price, and repair costs (if needed) <u>Wait time:</u> 30-60 days, until the car is resold (average time until car is sold)	DECISIONS  Predictions are turned into value by converting inputs (resale price, time-to-sell, risk, current market, etc.) into a recommended maximum purchase price and a simple buy/pass decision. The system applies user-defined constraints like target profit margin and holding cost to determine whether a car listing is worth acquiring. This process is controlled by adjustable parameters such as profit margin, risk tolerance, and holding period, which let the user shift recommendations between more aggressive (higher inventory, lower margin) and more conservative (fewer, higher-margin deals) strategies.	VALUE PROPOSITION  The end user is a used car dealer who uses the IDSS to decide which vehicles to acquire and at what price. Their objectives are to purchase vehicles at optimal prices, maintain high inventory turnover, and maximize profit while minimizing capital tied up in slow-selling or overvalued cars. They benefit from the system through data-driven predictions of resale value, expected days-to-sell, and profit over a typical 30-60 day resale period, which helps them make faster, more consistent, and lower-risk buying decisions. The workflow is integrated into a simple interface where the user inputs vehicle details and adjusts assumptions such as profit margin and holding costs, and the system then outputs a recommended maximum purchase price to support buy or pass decisions and increase average profit per car acquired, improve inventory turnover rate, and maximize return on return on invested capital.	DATA COLLECTION  <u>Initial set:</u> ~426k historical Craigslist listings (~300k after cleaning). Split chronologically by posting_date, 80% train / 20% test. <u>Continuous update:</u> pull live comparable listings from the MarketCheck API at each appraisal plus a weekly batch; retrain monthly on a rolling 18-24 month window. <u>Holdout:</u> the most recent listings and newly acquired cars are held out so we test on genuinely unseen, in-production vehicles. <u>Cost/constraints:</u> true labels (actual resale price and days-to-sell) only appear after the car is reconditioned and resold, a 30-60 day lag with real capital tied up. Listing prices are asking prices, corrected once the dealer's realized sale data comes in.	DATA SOURCES  <u>Historical Data Source:</u> Kaggle dataset: "Craigslist Cars & Trucks" The dataset has about 400 000 rows of historical data that can be used to train the model. <u>Live Data Source:</u> MarketCheck API This api aggregates over 5 billion vehicle listings since 2015 in real time and shows inventory across dealer, private-seller, and auction channels, along with market-analytics endpoints for average price, mileage, days-on-market, and Market Days Supply by make, model, and region.
IMPACT SIMULATION  The model can be deployed as the dealership should have all predictor values at the time of purchase. The Craigslist dataset will be split into training and testing sets to evaluate the IDSS' prediction performance. Additionally, the decisions will be evaluated on sale price and overall profit. Generally, the cost of overestimating a vehicle is higher than underestimating it. Overestimation may cause the dealership to purchase vehicles that won't sell for profit, whereas underestimation would likely be a missed opportunity. Since the model helps users decide whether or not to purchase a vehicle, the model should avoid over or under evaluating vehicles from certain regions.	MAKING PREDICTIONS  Predications are made real-time when a dealership inputs a vehicle listing. The system immediately estimates resale value, expected profit, and the final purchase or not recommendation, since vehicle purchases are typically done individually. The model may be trained on batches as the Craigslist dataset is re-scraped monthly. The target response time for the model is less than 5 seconds. Since the system is fairly simple, featurization (ie. vehicle age and post-processing (ie. profit estimate) can be completed in a couple of seconds. The targets are a simple buy/pass recommendation and expected profit.	USER INTERFACE  <u>Insights Needed:</u> The user needs a clear buy/pass recommendation, supported by recommended maximum purchase price, expected resale value, profit margin, and predicted time-to-sell (30-60 days) to assess profitability and inventory turnover risk. <u>Metrics and visualizations:</u> These are presented through a vehicle summary card (buy/pass and key metrics), a price comparison view (listing vs max purchase vs resale price), and an inventory table to compare profit, risk, and time-to-sell across multiple listings. <u>Adjustable parameters:</u> Users can control target profit margin, holding period, and risk tolerance. <u>User Inputs:</u> Inputs include manual entry of vehicle details or bulk upload of listings, along with assumptions like holding costs or repair estimates).	BUILDING MODELS  3 different prediction models will be in place. Resale Price Prediction Model Predicts the expected resale value of the car within the targeted market. Features include, make, model, year, mileage, condition and location. Days-Til-Sold Prediction Model Estimates the length of time a car will stay in inventory before it is sold. Helps account for holding costs and inventory turnover. Profitability Classification Model Predicts the probability that a car will meet the dealer's profit margin within the holding period. Outputs a Buy/Pass recommendation based on the user's risk tolerance. Models should be retrained monthly or whenever significant market shifts occur. The model will function as a RL model to the extent that real sales outcomes affect future predictions.	FEATURES  Input representations available at prediction time, from raw sources. <u>From kaggle and api:</u> Year/vehicle age, odometer, manufacturer, model, condition, title status, transmission, drive, cylinders, fuel, body type, paint_color, region/state, and posting season. <u>Engineered input:</u> age = current year - model year, mileage-per-year, brand reliability tier, regional price index, and seasonal demand flag. All are knowable before acquisition. <u>Adjustable Parameters:</u> profit margin, risk tolerance, and holding cost.

MONITORING

The accuracy of the ML system will be evaluated using both model performance metrics and business impact metrics.

1. Model Performance Metrics: Mean Absolute Error (MAE) for resale price predictions, Mean Absolute Error (MAE) for days-til-sold predictions, Precision, Recall and F1 Score for buy/pass recommendations and Calibration of predicted profit probabilities
2. Business Impact Metrics: Average Profit per car acquired, Inventory Turnover Rate, Percentage of purchased vehicles sold within the target 30-60 day window, Return on invested capital (ROIC), Reduction in losses from overpaying for vehicles and Increase in successful purchases meeting target profit margins

The system will continuously compare predicted resale prices, sales timeline's, and profits against actual outcomes after vehicles are sold. Significant degradation in prediction accuracy or profit performance will trigger model retraining and an investigation into data drift, market changes, or feature engineering issues. Dealer feedback on recommendations can also be collected to improve future model performance and user trust.

Expert Consultation

Name: Keith Fernandez (Data Engineer at Amazon Demand Forecasting)

We consulted Keith, a mentor from one of our co-op experiences at Amazon who works on the Demand Forecasting team, who helped us reframe the problem from a pure prediction task into a decision-support system focused on inventory, risk, and timing constraints. He emphasized that the value of forecasting comes from how predictions drive operational decisions, which shaped our shift toward a buy/pass framework and the use of real transaction-aligned data (Craigslist + market API) to reduce mismatch between training and real-world outcomes. This also influenced a modular system design separating price, time-to-sell, and decision logic to improve interpretability and decision quality.